AML Knowledge Library - Comprehensive Red Flags Master Document

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Overview

This master document consolidates **147+ red flags and indicators** from 8 authoritative AML/ATF sources. Each red flag includes:

- **Indicator Description**: What to look for
- **Data Signal**: What patterns in data indicate this
- **Feature Mapping**: Suggested features for model development
- **Source Reference**: Original document reference (see Source Documents section for full citations)

Source Reference Format: Each red flag includes a source reference in the format:

- `FINTRAC 18/19-SIDEL-025, Red Flag X` = FINTRAC Operational Alert reference number and red flag number
- `FinCEN FIN-2025-Alert001, Red Flag X` = FinCEN Alert reference number and red flag number
- `FinCEN FIN-2025-A003, Red Flag X` = FinCEN Advisory reference number and red flag number
- `FINTRAC ISO-OSI, Red Flag X` = FINTRAC Operational Alert (synthetic opioids)
- `FINTRAC ML-REC, Red Flag X` = FINTRAC Operational Alert (underground banking)
- `FINTRAC OAI-HTS-2021, Red Flag X` = FINTRAC Operational Alert (human trafficking)
- `FINTRAC Guidance, Red Flag X` = FINTRAC Guidance document

Purpose:

- Quick reference for all AML red flags in one place
- Feature engineering guide for Task 2
- Explanation reference for Task 3
- Comprehensive indicator library

Note: For detailed context, methodology, and additional information, refer to the individual source documents (01-08).

How to Use This Document

For Feature Engineering (Task 2)

1. Review red flags relevant to your data
2. Check "Data Signal" and "Feature Mapping" columns
3. Engineer features based on suggested mappings
4. Reference individual documents for detailed context

For Model Explainability (Task 3)

1. When a customer is flagged, identify which red flags match
2. Use "Indicator Description" and "Context" to explain why
3. Reference source documents for authoritative backing

For Risk Assessment

1. Review red flags by category
2. Note "High-Risk Combinations" sections in individual documents
3. Combine multiple indicators for stronger signals

Red Flags by Category

1. Professional Money Laundering (Trade & MSB)

Source: FINTRAC Operational Alert 18/19-SIDEL-025
Document: `01_Professional_Money_Laundering_Trade_MSB.md`

#	Red Flag	Data Signal	Feature Mapping	Source Reference
1	Phantom Shipments	Large EFT to trading companies, no goods movement	`volume_eft_to_trading_companies`, `txn_count_without_verification`	FINTRAC 18/19-SIDEL-025, Red Flag 1
2	Falsely Described Goods	Invoice amounts inconsistent with typical prices	`amount_vs_typical_price_ratio`, `merchant_category_anomalies`	FINTRAC 18/19-SIDEL-025, Red Flag 2
3	Multiple Invoicing	Multiple payments for same invoice	`duplicate_invoice_payments`, `payment_frequency_per_invoice`	FINTRAC 18/19-SIDEL-025, Red Flag 3
4	Over/Under Invoicing	Amounts significantly different from market prices	`amount_vs_market_price_deviation`, `over_invoice_flag`, `under_invoice_flag`	FINTRAC 18/19-SIDEL-025, Red Flag 4

#	Red Flag	Data Signal	Feature Mapping	Source Reference
5	Black Market Peso Exchange	EFT from Latin America to Canadian trading companies	`volume_eft_from_high_risk_countries`, `txn_count_to_trading_companies`	FINTRAC 18/19-SIDEL-025, Scheme Type 2
6	Canadian Trading Company Route	EFT from Latin America/U.S. to Canadian trading companies	`volume_eft_from_latin_america`, `high_risk_geography_flag`	FINTRAC 18/19-SIDEL-025, Red Flag 4
7	Correspondent Bank Route	EFT flows through Canadian banks to/from high-risk jurisdictions	`correspondent_bank_flows`, `high_risk_jurisdiction_transfers`	FINTRAC 18/19-SIDEL-025, Red Flag 4
8	High-Risk Trade Sector	Business in agri-food, textiles, electronics, toys, lumber, automotive	`industry_code_high_risk_trade`, `business_type_trading_company`	FINTRAC 18/19-SIDEL-025, Red Flag 1
9	Unusual Business Activities	Business activities outside norm or difficult to confirm	`business_activity_anomaly`, `missing_business_info_flag`	FINTRAC 18/19-SIDEL-025, Red Flag 2
10	Sudden Large Inflows	Sudden increase in EFT volume or count	`volume_eft_sudden_increase`, `txn_count_eft_spike`	FINTRAC 18/19-SIDEL-025, Red Flag 3
11	Geographic: China/Hong Kong	EFT to China/Hong Kong, from U.S./Latin America	`volume_eft_to_china_hk`, `volume_eft_from_latin_america`	FINTRAC 18/19-SIDEL-025, Red Flag 4
12	Geographic: U.S./Mexico/Latin America	EFT to/from U.S., Mexico, Latin America	`volume_eft_to_us_mexico_latam`, `cross_border_flows`	FINTRAC 18/19-SIDEL-025, Red Flag 5
13	Offshore Financial Centers	Transactions to/from Latvia, Cyprus, BVI, Panama, etc.	`offshore_financial_center_flag`, `high_risk_jurisdiction_count`	FINTRAC 18/19-SIDEL-025, Red Flag 6
14	Round Figure Payments	Payments in round figures or $50K increments	`round_amount_flag`, `just_below_threshold_count`, `amount_modulo_50000`	FINTRAC 18/19-SIDEL-025, Red Flag 7

#	Red Flag	Data Signal	Feature Mapping	Source Reference
15	UAE to Canada Transfers	EFT from UAE to Canada	`volume_eft_from_uae`, `uae_to_canada_flag`	FINTRAC 18/19-SIDEL-025, Red Flag 8
16	Flow-Through Activity	Money taken out as quickly as it flows in	`flow_through_velocity`, `time_between_deposit_withdrawal`	FINTRAC 18/19-SIDEL-025, Red Flag 9
17	Currency Import	Imports currency (USD) from Latin America	`cash_deposits_from_latam`, `usd_import_flag`	FINTRAC 18/19-SIDEL-025, Red Flag 10
18	Credit Card Overpayments	Large business purchases by credit card, funded by overpayments	`credit_card_overpayment_flag`, `large_business_purchase_cc`	FINTRAC 18/19-SIDEL-025, Red Flag 11
19	Legal Professional Accounts	Trade payments through legal professional accounts	`legal_professional_account_usage`, `trade_payment_through_lawyer`	FINTRAC 18/19-SIDEL-025, Red Flag 12
20	MSB Sudden Inflows/Outflows	Sudden large EFT/cash in, followed by multiple outflows	`sudden_inflow_outflow_pattern`, `multiple_unrelated_party_payments`	FINTRAC 18/19-SIDEL-025, Red Flag 13
21	MSB Currency Exchanges	Numerous CAD/USD/EUR exchanges	`currency_exchange_frequency`, `cad_usd_eur_exchanges`	FINTRAC 18/19-SIDEL-025, Red Flag 14
22	MSB High-Risk Country Business	Business with Iran, UAE, Kuwait, Hong Kong, China	`sanctioned_country_transactions`, `high_risk_country_business_pct`	FINTRAC 18/19-SIDEL-025, Red Flag 15
23	MSB Foreign Exchange Transfers	EFT from foreign exchange/trading companies for real estate/loans	`eft_from_fx_trading_companies`, `real_estate_loan_investment_flows`	FINTRAC 18/19-SIDEL-025, Red Flag 16

#	Red Flag	Data Signal	Feature Mapping	Source Reference
24	Personal Account MSB Activity	Personal account showing MSB-like patterns	`personal_account_business_activity`, `msb_like_personal_account`	FINTRAC 18/19-SIDEL-025, Red Flag 17
25	Multiple Identities	Multiple occupations, addresses, telephone numbers	`multiple_occupations_flag`, `multiple_addresses_flag`	FINTRAC 18/19-SIDEL-025, Red Flag 19
26	Living Beyond Means	Transaction volume exceeds declared income	`amount_to_income_ratio`, `lifestyle_mismatch`	FINTRAC 18/19-SIDEL-025, Red Flag 21

2. Bulk Cash Smuggling (Mexico TCOs)

Source: FinCEN Alert FIN-2025-Alert001
Document: `02_Bulk_Cash_Smuggling_Mexico_TCOs.md`

#	Red Flag	Data Signal	Feature Mapping	Source Reference
1	Structured Cash Deposits	Multiple cash deposits $9K-$10K in quick succession	`structured_cash_deposits_count`, `just_below_threshold_cash_deposits`	FinCEN FIN-2025-Alert001, Red Flag 1
2	Cross-Border Cash Movement	Large cash withdrawals before travel to Mexico	`large_cash_withdrawal_flag`, `border_region_cash_activity`	FinCEN FIN-2025-Alert001, Red Flag 2
3	Multiple Location Cash Deposits	Same person, multiple locations, same day	`multiple_location_cash_deposits_same_day`, `cash_deposit_location_diversity`	FinCEN FIN-2025-Alert001, Red Flag 3
4	Third-Party Cash Deposits	Cash deposits into accounts of others, especially Mexico-linked	`third_party_cash_deposit_flag`, `mexico_linked_account_deposits`	FinCEN FIN-2025-Alert001, Red Flag 4
5	Wire Transfers to Mexico After Cash	Large wires to Mexico following cash deposits	`wire_to_mexico_after_cash`, `cash_to_wire_repatriation_pattern`	FinCEN FIN-2025-Alert001, Red Flag 5

#	Red Flag	Data Signal	Feature Mapping	Source Reference
6	MSB Usage for Repatriation	MSB transfers to Mexico after cash deposits	`msb_to_mexico_transfers`, `western_union_mexico_flag`	FinCEN FIN-2025-Alert001, Red Flag 6
7	Money Orders/Bank Drafts	Money orders/drafts to Mexico, funded by cash	`money_order_purchase_to_mexico`, `cash_funded_money_orders`	FinCEN FIN-2025-Alert001, Red Flag 7
8	Rapid Cash Accumulation	Sudden cash spike followed by immediate transfers	`rapid_cash_accumulation`, `cash_to_transfer_velocity`	FinCEN FIN-2025-Alert001, Red Flag 8
9	Smuggling Corridors	Transactions in border regions or known corridors	`border_region_transaction_flag`, `smuggling_corridor_activity`	FinCEN FIN-2025-Alert001, Red Flag 9
10	Layered Transactions	Cash → multiple accounts → Mexico	`layered_transaction_count`, `multi_account_chain_flag`	FinCEN FIN-2025-Alert001, Red Flag 10
11	Mexico-U.S. Border Activity	High volume in border cities (El Paso, Laredo, etc.)	`border_region_transaction_count`, `border_city_activity_flag`	FinCEN FIN-2025-Alert001, Red Flag 11
12	Mexico-Based Entity Transactions	Frequent transactions with Mexico-based entities	`mexico_transaction_volume`, `mexico_entity_transaction_count`	FinCEN FIN-2025-Alert001, Red Flag 12
13	Lifestyle Inconsistencies	Low income, large cash deposits, Mexico connections	`income_transaction_mismatch_mexico`, `low_income_high_volume_mexico`	FinCEN FIN-2025-Alert001, Red Flag 13
14	Avoidance of Reporting Thresholds	Consistent just-below-threshold transactions	`just_below_ctr_threshold_count`, `structuring_pattern_mexico`	FinCEN FIN-2025-Alert001, Red Flag 14

#	Red Flag	Data Signal	Feature Mapping	Source Reference
15	Rapid Account Turnover	Accounts: cash in → immediate transfer out	`account_turnover_rate`, `cash_deposit_to_transfer_time`	FinCEN FIN-2025-Alert001, Red Flag 15

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3. Oil Smuggling (Mexico Cartels)

Source: FinCEN Alert FIN-2025-Alert002
Document: `03_Oil_Smuggling_Mexico_Cartels.md`

#	Red Flag	Data Signal	Feature Mapping	Source Reference
1	Payments to Oil Companies	Large payments to oil/petroleum companies, border regions	`oil_company_payment_volume`, `petroleum_merchant_transactions`	FinCEN FIN-2025-Alert002, Red Flag 1
2	Southwest Border Oil Transactions	Oil transactions in TX, NM, AZ, CA border states	`southwest_border_oil_activity`, `border_state_oil_transactions`	FinCEN FIN-2025-Alert002, Red Flag 2
3	Unusual Payment Methods	Large cash or structured payments for oil	`cash_oil_payments`, `structured_oil_payments`	FinCEN FIN-2025-Alert002, Red Flag 3
4	Rapid Oil Transaction Sequences	High frequency oil transactions in short time	`oil_transaction_velocity`, `rapid_oil_transaction_sequence`	FinCEN FIN-2025-Alert002, Red Flag 4
5	Cross-Border Oil Flows	Oil transactions across U.S.-Mexico border	`cross_border_oil_flows`, `mexico_oil_transactions`	FinCEN FIN-2025-Alert002, Red Flag 5
6	Layered Oil Transactions	Complex chains through multiple oil businesses	`oil_business_transaction_chain`, `layered_oil_transactions`	FinCEN FIN-2025-Alert002, Red Flag 6
7	Border Region Fuel Activity	High volume in border cities (El Paso, Laredo, San Diego)	`border_city_oil_activity`, `fuel_transaction_border_cities`	FinCEN FIN-2025-Alert002, Red Flag 7

#	Red Flag	Data Signal	Feature Mapping	Source Reference
8	Transactions Near Oil Infrastructure	Transactions near pipelines, refineries, storage	`oil_infrastructure_proximity_flag`, `pipeline_region_transactions`	FinCEN FIN-2025-Alert002, Red Flag 8
9	Oil Business Unusual Activity	Oil business with patterns inconsistent with operations	`oil_business_anomaly_flag`, `unusual_oil_business_activity`	FinCEN FIN-2025-Alert002, Red Flag 9
10	New Oil Business High Volume	Recently established oil business with immediate high volume	`new_oil_business_high_volume`, `recent_oil_business_flag`	FinCEN FIN-2025-Alert002, Red Flag 10
11	Round-Number Oil Transactions	Oil transactions in round numbers ($50K, $100K)	`round_oil_transaction_flag`, `bulk_oil_sale_indicator`	FinCEN FIN-2025-Alert002, Red Flag 11
12	Large-Value Oil Transactions	Exceptionally large oil/fuel purchases	`large_oil_transaction_flag`, `excessive_oil_purchase_amount`	FinCEN FIN-2025-Alert002, Red Flag 12
13	Oil + Drug Proceeds Patterns	Oil transactions + cash activity + structuring	`oil_drug_proceeds_combination`, `multi_criminal_activity_flag`	FinCEN FIN-2025-Alert002, Red Flag 13
14	Oil Business Cartel Links	Oil business transacting with cartel-linked accounts	`oil_cartel_linked_transactions`, `cartel_network_oil_flag`	FinCEN FIN-2025-Alert002, Red Flag 14

4. Chinese ML Networks (Mexico TCOs)

Source: FinCEN Advisory FIN-2025-A003
Document: `04_Chinese_ML_Networks_Mexico_TCOs.md`

#	Red Flag	Data Signal	Feature Mapping	Source Reference
1	China/Hong Kong Transactions	Large/frequent EFT to China/Hong Kong	`volume_eft_to_china_hk`, `txn_count_to_china_hk`	FinCEN FIN-2025-A003, Red Flag 1

#	Red Flag	Data Signal	Feature Mapping	Source Reference
2	Chinese Network Intermediaries	Transactions through Chinese ML network accounts	`chinese_network_intermediary_flag`, `chinese_ml_network_account_transactions`	FinCEN FIN-2025-A003, Red Flag 2
3	Rapid China-Bound Sequences	Rapid transaction sequences ending in China/HK	`rapid_china_bound_sequence`, `china_transfer_velocity`	FinCEN FIN-2025-A003, Red Flag 3
4	Mexico-to-China Flows	EFT from Mexico-based entities to China/HK	`mexico_to_china_flows`, `mexico_china_network_flow_flag`	FinCEN FIN-2025-A003, Red Flag 4
5	Layered Through Chinese Networks	Complex chains through Chinese intermediaries	`chinese_network_layering_count`, `layered_china_transactions`	FinCEN FIN-2025-A003, Red Flag 5
6	U.S.-China-Mexico Triangle	Transactions: U.S. → China → Mexico (or vice versa)	`us_china_mexico_triangle_flag`, `triangle_transaction_chain`	FinCEN FIN-2025-A003, Red Flag 6
7	Currency Exchange Through Chinese Networks	Exchanges involving CNY or HKD	`chinese_yuan_exchange_flag`, `hkd_exchange_volume`	FinCEN FIN-2025-A003, Red Flag 7
8	Multi-Currency Transactions	Transactions: USD, CNY, MXN with China connections	`multi_currency_china_flag`, `usd_cny_mxn_transactions`	FinCEN FIN-2025-A003, Red Flag 8
9	Chinese-Owned Business TCO Links	Chinese businesses transacting with TCO-linked accounts	`chinese_business_tco_connections`, `china_mexico_business_links`	FinCEN FIN-2025-A003, Red Flag 9
10	Import/Export China-Mexico Trade	Import/export companies with China-Mexico trade, unusual patterns	`china_mexico_trade_company_flag`, `trade_based_ml_china_mexico`	FinCEN FIN-2025-A003, Red Flag 10
11	Chinese Financial Hubs	Transactions: Hong Kong, Shanghai, Beijing, Shenzhen	`chinese_financial_hub_transactions`, `hong_kong_transaction_flag`	FinCEN FIN-2025-A003, Red Flag 11

#	Red Flag	Data Signal	Feature Mapping	Source Reference
12	Offshore Centers + Chinese Connections	Offshore centers (BVI, Cayman) with China connections	`offshore_china_connections`, `chinese_offshore_network_flag`	FinCEN FIN-2025-A003, Red Flag 12
13	Structured Transactions to China	Multiple just-below-$3K transactions to China/HK	`structured_china_transfers`, `china_structuring_pattern`	FinCEN FIN-2025-A003, Red Flag 13
14	Round-Number Transfers to China	Transfers to China/HK in round numbers	`round_china_transfers`, `bulk_china_transfer_indicator`	FinCEN FIN-2025-A003, Red Flag 14
15	Cash-to-China Transfer Patterns	Large cash deposits → wire transfers to China/HK	`cash_to_china_transfer_pattern`, `drug_proceeds_china_flow`	FinCEN FIN-2025-A003, Red Flag 15
16	Mexico TCO Proceeds Through Chinese Networks	Mexico TCO drug proceeds laundered through Chinese networks	`mexico_tco_china_network_flag`, `tco_china_collaboration_indicator`	FinCEN FIN-2025-A003, Red Flag 16

5. Synthetic Opioids Proceeds

Source: FINTRAC Operational Alert (ISO-OSI)
Document: `05_Synthetic_Opioids_Proceeds.md`

#	Red Flag	Data Signal	Feature Mapping	Source Reference
1	Large Cash Deposits	Large/frequent cash deposits, especially structured	`large_cash_deposit_count`, `structured_cash_deposits`	FINTRAC ISO-OSI, Red Flag 1
2	Rapid Cash Accumulation	Sudden increase in cash deposits	`rapid_cash_accumulation`, `sudden_cash_spike_flag`	FINTRAC ISO-OSI, Red Flag 2
3	Multiple Location Cash Deposits	Same person, multiple locations, same day	`multiple_location_cash_deposits`, `cash_deposit_location_diversity`	FINTRAC ISO-OSI, Red Flag 3
4	Cash-Intensive Businesses	Cash businesses (restaurants, retail) with unusually high cash	`cash_intensive_business_flag`, `unusual_cash_business_volume`	FINTRAC ISO-OSI, Red Flag 4

#	Red Flag	Data Signal	Feature Mapping	Source Reference
5	New Business High Cash	Recently established business with immediate high cash	`new_business_high_cash`, `recent_business_cash_flag`	FINTRAC ISO-OSI, Red Flag 5
6	No Legitimate Operations	Business with cash deposits but no business activity	`front_business_flag`, `no_legitimate_operations`	FINTRAC ISO-OSI, Red Flag 6
7	Rapid Money Movement	Cash deposits → immediate wire transfers/EFTs	`rapid_money_movement`, `cash_to_transfer_velocity`	FINTRAC ISO-OSI, Red Flag 7
8	High Transaction Frequency	Unusually high number of transactions, especially cash	`high_transaction_frequency`, `cash_transaction_frequency`	FINTRAC ISO-OSI, Red Flag 8
9	High-Risk Area Transactions	Transactions in areas known for opioid trafficking	`high_risk_area_transactions`, `opioid_epidemic_region_flag`	FINTRAC ISO-OSI, Red Flag 9
10	Cross-Border Patterns	Transactions with countries known for opioid production	`cross_border_opioid_flows`, `high_risk_country_transactions`	FINTRAC ISO-OSI, Red Flag 10
11	Just Below Threshold	Multiple transactions just below $10K	`just_below_threshold_count`, `structuring_pattern_flag`	FINTRAC ISO-OSI, Red Flag 11
12	Round-Number Transactions	High percentage of round-number transactions	`round_amount_pct`, `round_number_transaction_count`	FINTRAC ISO-OSI, Red Flag 12
13	Lifestyle Inconsistencies	Low income, large cash deposits, high-value purchases	`lifestyle_inconsistency_flag`, `income_transaction_mismatch`	FINTRAC ISO-OSI, Red Flag 13
14	High-Value Cash Purchases	Large purchases (vehicles, real estate) funded by cash	`high_value_cash_purchase_flag`, `luxury_asset_cash_funding`	FINTRAC ISO-OSI, Red Flag 14
15	Multiple Related Accounts	Network of related accounts with similar cash patterns	`related_account_network_flag`, `network_cash_pattern_similarity`	FINTRAC ISO-OSI, Red Flag 15
16	Third-Party Cash Deposits	Multiple individuals depositing into same account	`third_party_cash_deposits`, `multiple_depositors_same_account`	FINTRAC ISO-OSI, Red Flag 16

6. Underground Banking Schemes

Source: FINTRAC Operational Alert (ML-REC)
Document: `06_Underground_Banking_Schemes.md`

#	Red Flag	Data Signal	Feature Mapping	Source Reference
1	Informal Value Transfer Indicators	Transactions suggesting hawala/hundi/IVTS	`hawala_indicator_flag`, `informal_value_transfer_pattern`	FINTRAC ML-REC, Red Flag 1
2	Transactions Without Business Purpose	Large transactions between unrelated parties	`unrelated_party_large_transactions`, `no_business_relationship_flag`	FINTRAC ML-REC, Red Flag 2
3	Rapid Cross-Border Value Movement	Value moves across borders without formal transactions	`cross_border_informal_transfer`, `underground_cross_border_flag`	FINTRAC ML-REC, Red Flag 3
4	Hawala Operator Patterns	Accounts: multiple unrelated sources → multiple destinations	`hawala_operator_pattern`, `intermediary_hawala_flag`	FINTRAC ML-REC, Red Flag 4
5	Settlement Through Alternative Means	Transactions settled through goods/services	`alternative_settlement_flag`, `trade_based_settlement`	FINTRAC ML-REC, Red Flag 5
6	Minimal Documentation	Large transactions with minimal documentation	`minimal_documentation_flag`, `undocumented_large_transactions`	FINTRAC ML-REC, Red Flag 6
7	High-Risk Country Connections	Transactions with countries known for underground banking	`underground_banking_country_flag`, `hawala_prevalent_region_transactions`	FINTRAC ML-REC, Red Flag 7
8	Unbanked Population Transactions	Transactions with unbanked communities	`unbanked_population_transactions`, `informal_financial_system_usage`	FINTRAC ML-REC, Red Flag 8
9	Asymmetric Transaction Patterns	Large in → many small out, or many small in → large out	`asymmetric_transaction_pattern`, `funnel_account_indicator`	FINTRAC ML-REC, Red Flag 9
10	Round-Number Transactions	Round numbers, especially foreign currency equivalents	`round_number_underground_transactions`, `hawala_round_amounts`	FINTRAC ML-REC, Red Flag 10
11	Unregistered MSB Activity	MSB-like activity without MSB registration	`unregistered_msb_activity`, `msb_like_underground_flag`	FINTRAC ML-REC, Red Flag 11
12	Multiple Currency Exchanges	Frequent currency exchanges, multiple currencies	`frequent_currency_exchanges`, `multi_currency_underground_flag`	FINTRAC ML-REC, Red Flag 12

#	Red Flag	Data Signal	Feature Mapping	Source Reference
13	Cash-to-Underground Banking	Large cash deposits → underground banking indicators	`cash_to_underground_banking`, `criminal_proceeds_underground_flag`	FINTRAC ML-REC, Red Flag 13
14	Layering Through Underground Banking	Complex chains through underground banking intermediaries	`underground_banking_layering`, `hawala_intermediary_chain`	FINTRAC ML-REC, Red Flag 14

7. Human Trafficking Proceeds

Source: FINTRAC Operational Alert (OAI-HTS-2021)
Document: `07_Human_Trafficking_Proceeds.md`

#	Red Flag	Data Signal	Feature Mapping	Source Reference
1	High-Volume Cash Deposits	Frequent cash deposits, often just below thresholds	`high_volume_cash_deposits`, `frequent_cash_deposit_pattern`	FINTRAC OAI-HTS-2021, Red Flag 1
2	Multiple Location Cash Deposits	Same person, multiple locations, same day	`multiple_location_cash_deposits_trafficking`, `same_day_multi_location_cash`	FINTRAC OAI-HTS-2021, Red Flag 2
3	Third-Party Cash Deposits	Multiple individuals depositing into trafficker account	`third_party_cash_deposits_trafficking`, `trafficker_account_deposits`	FINTRAC OAI-HTS-2021, Red Flag 3
4	Cash-Intensive Trafficking Businesses	Massage parlors, escort services with high cash	`cash_intensive_trafficking_business`, `massage_parlor_high_cash`	FINTRAC OAI-HTS-2021, Red Flag 4
5	Online Advertisement Revenue	Revenue from escort websites, adult services	`online_adult_service_revenue`, `escort_website_transactions`	FINTRAC OAI-HTS-2021, Red Flag 5
6	Hotel/Motel Transactions	Frequent hotel/motel transactions, especially cash	`hotel_motel_transaction_frequency`, `cash_hotel_payments`	FINTRAC OAI-HTS-2021, Red Flag 6

#	Red Flag	Data Signal	Feature Mapping	Source Reference
7	Rapid Cash-to-Asset Conversion	Cash deposits → immediate asset purchases	`rapid_cash_to_asset_conversion`, `trafficking_asset_integration`	FINTRAC OAI-HTS-2021, Red Flag 7
8	Structured Transactions	Multiple just-below-$10K transactions	`structured_trafficking_transactions`, `just_below_threshold_trafficking`	FINTRAC OAI-HTS-2021, Red Flag 8
9	Round-Number Transactions	High percentage of round-number transactions	`round_number_trafficking_transactions`, `round_service_payments`	FINTRAC OAI-HTS-2021, Red Flag 9
10	High-Risk Area Transactions	Transactions in known trafficking areas	`high_risk_trafficking_area_transactions`, `trafficking_hub_activity`	FINTRAC OAI-HTS-2021, Red Flag 10
11	Cross-Border Patterns	Transactions with source countries for victims	`cross_border_trafficking_flows`, `source_country_transactions`	FINTRAC OAI-HTS-2021, Red Flag 11
12	Control Over Victim Accounts	One person controlling multiple accounts (victims)	`account_control_flag`, `multiple_account_control`	FINTRAC OAI-HTS-2021, Red Flag 12
13	Lifestyle Inconsistencies	Low income, large cash deposits, high-value purchases	`lifestyle_inconsistency_trafficking`, `low_income_high_volume_trafficking`	FINTRAC OAI-HTS-2021, Red Flag 13
14	High-Value Purchases	Large purchases (vehicles, jewelry, real estate) with cash	`high_value_trafficking_purchases`, `luxury_asset_trafficking_funding`	FINTRAC OAI-HTS-2021, Red Flag 14
15	Network of Related Accounts	Related accounts with similar cash/transaction patterns	`trafficking_network_accounts`, `network_cash_similarity_trafficking`	FINTRAC OAI-HTS-2021, Red Flag 15
16	MSB Usage	MSB transfers, especially to source countries	`msb_trafficking_usage`, `western_union_trafficking_flag`	FINTRAC OAI-HTS-2021, Red Flag 16

#	Red Flag	Data Signal	Feature Mapping	Source Reference
17	Unusual Transaction Timing	Transactions at unusual times (late night, early morning)	`unusual_timing_trafficking`, `late_night_trafficking_transactions`	FINTRAC OAI-HTS-2021, Red Flag 17
18	Weekend/Peak Period Activity	High activity during weekends or peak periods	`weekend_trafficking_activity`, `peak_period_transactions`	FINTRAC OAI-HTS-2021, Red Flag 18

8. General ML/TF Indicators

Source: FINTRAC Guidance (Financial Entities)
Document: `08_General_ML_TF_Indicators.md`

#	Red Flag	Data Signal	Feature Mapping	Source Reference
1	Unusual Transaction Patterns	Transactions unusual for customer profile	`transaction_pattern_anomaly`, `unusual_customer_behavior`	FINTRAC Guidance, Red Flag 1
2	Rapid Movement of Funds	Funds moving quickly through accounts	`rapid_fund_movement`, `high_velocity_transactions`	FINTRAC Guidance, Red Flag 2
3	Complex Transaction Structures	Complex structures: multiple parties/accounts/jurisdictions	`complex_transaction_structure`, `layered_transaction_count`	FINTRAC Guidance, Red Flag 3
4	Round-Number Transactions	High frequency of round-number transactions	`round_number_transaction_pct`, `bulk_transfer_indicator`	FINTRAC Guidance, Red Flag 4
5	Just Below Threshold	Multiple just-below-$10K or $3K transactions	`just_below_threshold_count`, `structuring_pattern_flag`	FINTRAC Guidance, Red Flag 5
6	Multiple Small Transactions	Breaking large amounts into small transactions	`multiple_small_transactions`, `fragmented_transaction_pattern`	FINTRAC Guidance, Red Flag 6
7	Same-Day Multi-Location	Same person, multiple locations, same day	`same_day_multi_location_transactions`, `distributed_transaction_pattern`	FINTRAC Guidance, Red Flag 7
8	Reluctance to Provide Information	Missing/incomplete customer information	`missing_customer_info_flag`, `information_reluctance_indicator`	FINTRAC Guidance, Red Flag 8

#	Red Flag	Data Signal	Feature Mapping	Source Reference
9	Unusual Account Activity	Activity inconsistent with customer profile	`account_activity_anomaly`, `profile_transaction_mismatch`	FINTRAC Guidance, Red Flag 9
10	Lifestyle Inconsistencies	Transactions beyond declared income/means	`lifestyle_inconsistency_flag`, `beyond_means_indicator`	FINTRAC Guidance, Red Flag 10
11	High-Risk Jurisdiction Transactions	Transactions with high-risk countries	`high_risk_jurisdiction_transactions`, `sanctioned_country_flag`	FINTRAC Guidance, Red Flag 11
12	Offshore Financial Centers	Transactions with offshore centers/tax havens	`offshore_financial_center_transactions`, `tax_haven_flag`	FINTRAC Guidance, Red Flag 12
13	Sanctioned Countries	Transactions with sanctioned countries	`sanctioned_country_transactions`, `sanctions_violation_flag`	FINTRAC Guidance, Red Flag 13
14	Multiple Accounts Similar Patterns	Multiple accounts showing similar unusual patterns	`multiple_accounts_similar_patterns`, `account_network_flag`	FINTRAC Guidance, Red Flag 14
15	Third-Party Account Usage	Account used primarily for third-party transactions	`third_party_account_usage`, `conduit_account_flag`	FINTRAC Guidance, Red Flag 15
16	Funnel Accounts	Many-to-one or one-to-many transaction patterns	`funnel_account_flag`, `aggregation_account_indicator`	FINTRAC Guidance, Red Flag 16
17	Large Cash Transactions	Unusually large cash deposits/withdrawals	`large_cash_transaction_flag`, `unusual_cash_volume`	FINTRAC Guidance, Red Flag 17
18	Frequent Cash Deposits	Frequent cash deposits, especially structured	`frequent_cash_deposits`, `structured_cash_deposit_pattern`	FINTRAC Guidance, Red Flag 18

#	Red Flag	Data Signal	Feature Mapping	Source Reference
19	Cash-Intensive Business Anomalies	Cash business with unusual patterns	`cash_intensive_business_anomaly`, `front_business_indicator`	FINTRAC Guidance, Red Flag 19
20	Rapid Wire Transfer Sequences	Rapid wire sequences, especially high-risk jurisdictions	`rapid_wire_sequence`, `wire_transfer_velocity`	FINTRAC Guidance, Red Flag 20
21	Wire Transfers Without Purpose	Large wires with no apparent business purpose	`wire_without_purpose_flag`, `unexplained_wire_transfers`	FINTRAC Guidance, Red Flag 21
22	International Wire Patterns	Unusual international wire patterns, high-risk countries	`international_wire_pattern_anomaly`, `cross_border_wire_activity`	FINTRAC Guidance, Red Flag 22
23	Small, Frequent Transactions	Multiple small transactions (terrorist financing)	`small_frequent_transactions`, `terrorist_financing_pattern`	FINTRAC Guidance, Red Flag 23
24	Charitable Organization Concerns	Transactions with charities, especially high-risk regions	`charitable_organization_transactions`, `charity_terrorist_financing_flag`	FINTRAC Guidance, Red Flag 24
25	Transactions to Conflict Zones	Transactions to conflict zones or terrorist activity regions	`conflict_zone_transactions`, `terrorist_activity_region_flag`	FINTRAC Guidance, Red Flag 25
26	Unusual Business Relationships	Transactions between unrelated businesses/individuals	`unusual_business_relationship`, `unrelated_party_transactions`	FINTRAC Guidance, Red Flag 26
27	Shell Company Indicators	Transactions with shell companies or minimal operations	`shell_company_transactions`, `minimal_operations_company_flag`	FINTRAC Guidance, Red Flag 27
28	Complex Ownership Structures	Business with complex/opaque ownership	`complex_ownership_structure`, `opaque_ownership_flag`	FINTRAC Guidance, Red Flag 28

Feature Mapping Quick Reference

High-Priority Features (Critical Red Flags)

Wire Transfer Features:

- `wire_txn_count`, `wire_volume_total`, `wire_volume_avg`, `has_wire_transfers`

Geographic Risk Features:

- `high_risk_country_transaction_count`, `offshore_financial_center_flag`, `sanctioned_country_transactions`

Cash Activity Features:

- `large_cash_deposit_count`, `structured_cash_deposits`, `multiple_location_cash_deposits_same_day`

Velocity Features:

- `rapid_fund_movement`, `flow_through_velocity`, `cash_to_transfer_velocity`

Structuring Features:

- `just_below_threshold_count`, `structuring_pattern_flag`, `round_amount_pct`

Feature Categories

1. **Transaction Amount Features**: `amount_mean`, `amount_max_ratio_mean`, `large_txn_count`
2. **Transaction Frequency Features**: `txn_count_total`, `txn_per_day_avg`, `high_frequency_customer_flag`
3. **Channel Features**: `wire_txn_count`, `abm_cash_txn_count`, `channels_used_count`
4. **Geographic Features**: `high_risk_country_transaction_count`, `country_diversity`, `border_region_activity_flag`
5. **Time-Based Features**: `after_hours_txn_pct`, `weekend_txn_pct`, `midnight_txn_count`
6. **Business Profile Features**: `industry_code_high_risk_trade`, `business_activity_anomaly`, `occupation_income_mismatch`
7. **Behavioral Features**: `lifestyle_inconsistency_flag`, `account_turnover_rate`, `funnel_account_flag`

Source Documents

For detailed context, methodology, and additional information, refer to:

1. **01_Professional_Money_Laundering_Trade_MSB.md**
   - **Source**: FINTRAC Operational Alert
   - **Reference**: 18/19-SIDEL-025
   - **Publication Date**: July 18, 2018
   - **URL**: https://fintrac-canafe.canada.ca/intel/operation/oai-ml-eng
2. **02_Bulk_Cash_Smuggling_Mexico_TCOs.md**
   - **Source**: FinCEN Alert
   - **Reference**: FIN-2025-Alert001
   - **URL**: https://www.fincen.gov/system/files/shared/BCS-Alert-FINAL-508C.pdf
3. **03_Oil_Smuggling_Mexico_Cartels.md**
   - **Source**: FinCEN Alert
   - **Reference**: FIN-2025-Alert002

    - **URL**: https://www.fincen.gov/system/files/shared/FinCEN-Alert-Oil-Smuggling-FINAL-508C.pdf

4. **04_Chinese_ML_Networks_Mexico_TCOs.md**
    - **Source**: FinCEN Advisory
    - **Reference**: FIN-2025-A003
    - **URL**: https://www.fincen.gov/resources/advisories/fincen-advisory-fin-2025-a003
5. **05_Synthetic_Opioids_Proceeds.md**
    - **Source**: FINTRAC Operational Alert
    - **Reference**: ISO-OSI (Illicit Synthetic Opioids)
    - **URL**: https://fintrac-canafe.canada.ca/intel/operation/iso-osi-eng
6. **06_Underground_Banking_Schemes.md**
    - **Source**: FINTRAC Operational Alert
    - **Reference**: ML-REC (Money Laundering - Underground Banking)
    - **URL**: https://fintrac-canafe.canada.ca/intel/operation/ml-rec-eng
7. **07_Human_Trafficking_Proceeds.md**
    - **Source**: FINTRAC Operational Alert
    - **Reference**: OAI-HTS-2021 (Human Trafficking for Sexual Exploitation)
    - **URL**: https://fintrac-canafe.canada.ca/intel/operation/oai-hts-2021-eng
8. **08_General_ML_TF_Indicators.md**
    - **Source**: FINTRAC Guidance
    - **Reference**: Financial Entities
    - **URL**: https://fintrac-canafe.canada.ca/guidance-directives/transaction-operation/indicators-indicateurs/fin_mltf-eng

Source Document Contents

All source documents include:

- **Detailed red flag descriptions** with full context and explanations
- **Risk assessment guidance** and high-risk combination indicators
- **Integration notes** for Tasks 2 (Detection Model) and 3 (Explainability)
- **Full source citations** with URLs, publication dates, and reference numbers
- **Methodology and background** information

Traceability

Every red flag in this master document:

- ☑ Has a source reference linking to the original document
- ☑ Can be traced back to the authoritative source
- ☑ Includes document reference number for verification
- ☑ Links to full source document for detailed context

Usage Tips

For Quick Reference

- Use the tables above to quickly identify relevant red flags
- Check "Data Signal" column to understand what to look for in data
- Use "Feature Mapping" column for feature engineering

For Detailed Analysis

- Refer to individual source documents (01-08) for:
  - Detailed context and explanations
  - Risk assessment frameworks
  - High-risk combination guidance
  - Source citations and traceability

For Feature Engineering

1. Review red flags relevant to your data
2. Check feature mapping suggestions
3. Engineer customer-level aggregations
4. Combine multiple features for stronger signals

For Model Explainability

1. Identify which red flags match flagged customers
2. Use indicator descriptions to explain decisions
3. Reference source documents for authoritative backing

Last Updated: 2025-01-XX
Total Red Flags: 147+
Source Documents: 8
Status: Complete and Comprehensive